DRUGS · ALLERGY, RHINITIS AND BIOLOGICS

Allergen immunotherapy

Allergen immunotherapy gives gradually increasing then maintenance doses of an allergen to induce tolerance. It is the only treatment that changes the natural history of allergic rhinitis: it reduces symptoms and medication use, can prevent new sensitisations and reduces the risk of children developing asthma, and its benefit persists for years after a 3-year course. It is given subcutaneously in specialist centres or as daily sublingual tablets or drops.

1 CHAPTER ONE At a glance

The treatment on one page.

1.1 Summary

ALLERGEN IMMUNOTHERAPY

	SUMMARY
Forms	Subcutaneous immunotherapy (SCIT): injections in a specialist clinic; sublingual immunotherapy (SLIT): daily tablets or drops at home, first dose under supervision
How it works	Induces regulatory T cells and IL-10 and TGF-beta, shifts responses from Th2 to Th1, raises allergen-specific IgG4 that blocks IgE, and reduces mast cell and eosinophil activity
ENT uses	Moderate to severe IgE-mediated allergic rhinitis (grass and tree pollen, house dust mite) with symptoms despite optimal medication, confirmed by history plus skin prick test or specific IgE
Course	Usually 3 years; SLIT pollen tablets started at least 2 to 4 months before the season and taken all year
Avoid	Severe or uncontrolled asthma (FEV1 under 70%); active autoimmune disease or malignancy; starting during pregnancy; beta blockers (relative, as anaphylaxis is harder to treat); poor adherence

2 CHAPTER TWO Using it in ENT

Selecting patients.

2.1 Who benefits

A patient with clearly seasonal or perennial symptoms that correlate with one or two relevant allergens, confirmed by testing, whose symptoms persist despite nasal steroids, antihistamines and allergen avoidance, and who will commit to 3 years. Polysensitised patients may still benefit if the clinically relevant allergen is clear.

2.2 SCIT versus SLIT

SCIT has a long track record but needs clinic visits and observation after each injection because of the risk of systemic reactions. SLIT is safer (local mouth itch and swelling are common, anaphylaxis rare) and convenient, but adherence falls with time.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- SCIT: local swelling; **systemic reactions and anaphylaxis**, so injections are given only where resuscitation is available, with 60 minutes of observation.
- SLIT: oral itch and swelling, throat irritation, gastrointestinal upset; eosinophilic oesophagitis rarely.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Indications and contraindications.
- Mechanism (IgG4 blocking antibodies, regulatory T cells).
- The disease-modifying effect and course length.
- Safety arrangements for SCIT.

4.2 Common pitfalls

COMMON PITFALLS

- Offering immunotherapy without confirming IgE sensitisation to the relevant allergen.
- Giving SCIT to a patient with uncontrolled asthma.

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